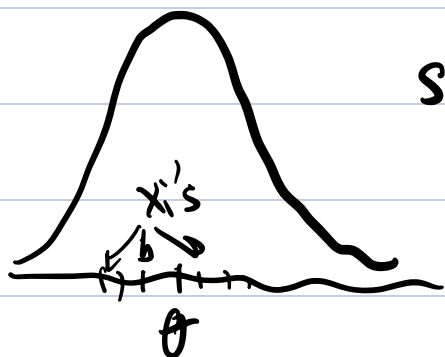


Suppose we observe $X_1, \dots, X_n$ iid f_θ

An estimator $\hat{\theta}_n$ of θ is any function of data:

$$\hat{\theta}_n = \theta(X_1, \dots, X_n)$$

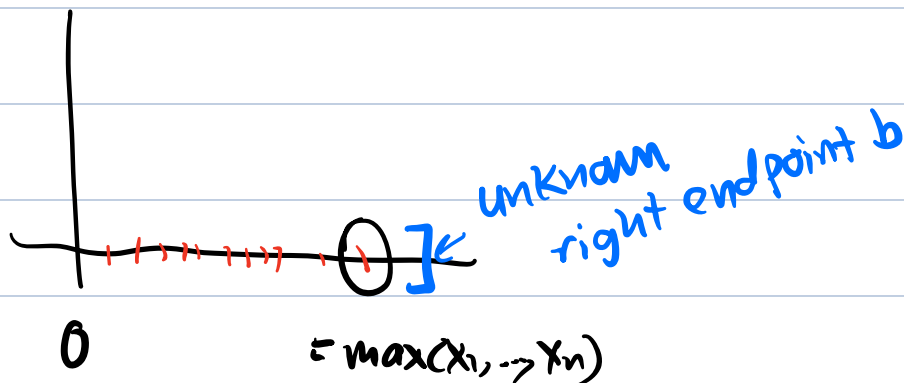
Examples $\hat{\theta}_n = \bar{X}_n$, good choice for $N(\theta, 1)$



Since θ is the expectation
(or "population mean")
of $N(\theta, 1)$

makes sense to estimate it by the sample mean

$\hat{\theta}_n = \max(X_1, \dots, X_n)$, good choice for $\text{Unif}(0, b)$



Bad estimators: $\hat{\theta}_n = X_1$, $\hat{\theta}_n = 4$

$\hat{\theta}_n$ is a random variable since $X_1, \dots, X_n$ are random.
 So we can study its distribution,
 called the **sampling distribution**

Example $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Bernoulli}(p)$
 (n coin flips), want to find p .

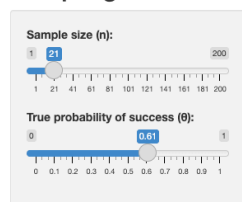
Via $\hat{p}_n = \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i = \text{proportion of heads}$

$$\bar{X}_n = \frac{1}{n} \times \text{Binomial}(n, p)$$

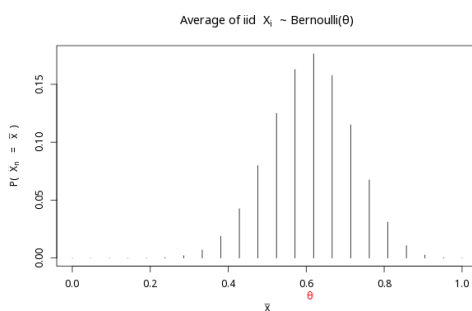
So $\text{Range}(\bar{X}_n) = \{0, \frac{1}{n}, \frac{2}{n}, \dots, \frac{n-1}{n}, 1\}$
 and

$$P(\bar{X}_n = \frac{k}{n}) = P(\underbrace{\sum_{i=1}^n X_i = k}_{\text{Binom}(n, p)}) = \underbrace{\binom{n}{k} p^k (1-p)^{n-k}}_{\text{Binom}(n, p)}$$

Sampling distribution of the sample proportion

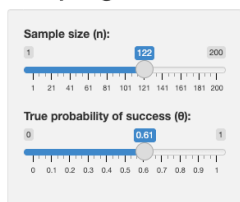


$n=21$

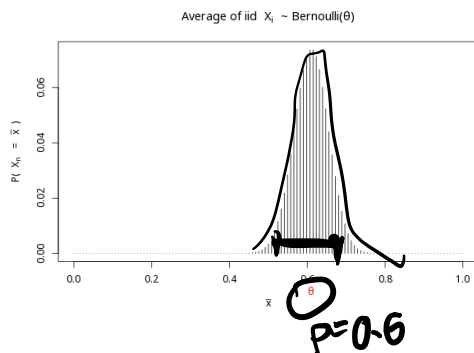


In general,
 may not be able
 to compute $\hat{\theta}_n$'s
 distribution

Sampling distribution of the sample proportion



$n=122$
 $p=0.6$



exactly.

But if $\hat{\theta}_n = \bar{X}_n$,
the CLT tells
us that

$\bar{X}_n$ is approx.
normally distributed.

(for n large enough)

Not true e.g. for $\max(X_1, \dots, X_n)$

Sampling dist. is important for
constructing confidence intervals

How good an estimator is:

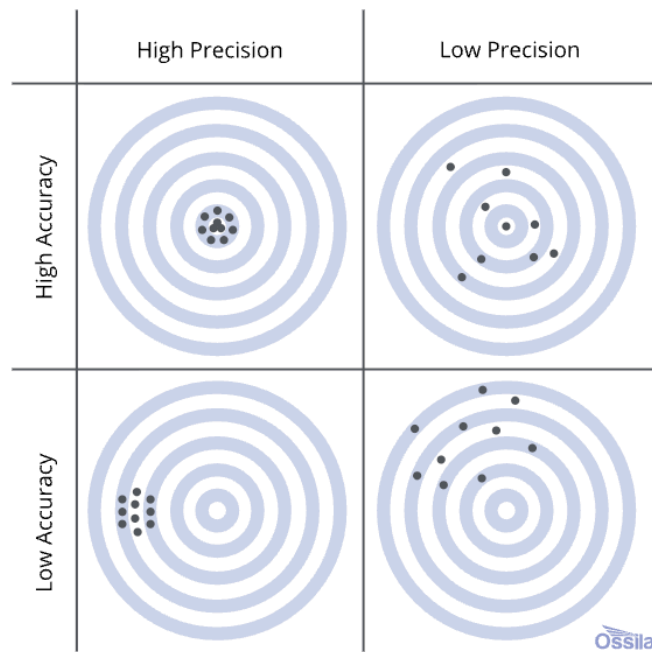
compute its bias and variance

Def Bias of $\hat{\theta}_n$ is

$$\text{bias}(\hat{\theta}_n) = E[\hat{\theta}_n] - \theta$$

want bias to be small

want variance to be small



high precision = low variance
high accuracy = low bias

Examples $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Ber}(p)$

Estimators: $\hat{p}_n = \frac{1}{2}$, $\hat{p}_n = X_1$, $\hat{p}_n = \bar{X}_n$

$\text{Bias}(\frac{1}{2}) = \frac{1}{2} - p$. Nails it when $p = \frac{1}{2}$
Bad when p is far from $\frac{1}{2}$

$\text{Var}(\frac{1}{2}) = 0$. Good!

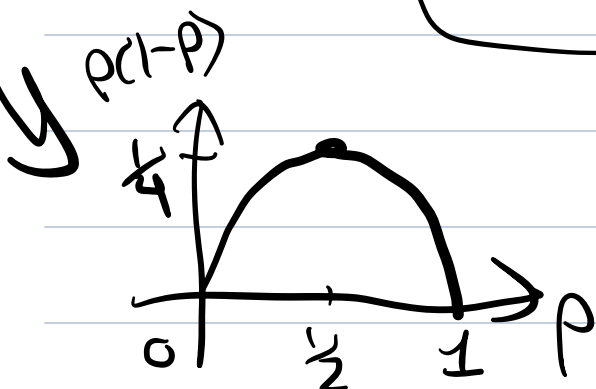
$\text{Bias}(X_1) = E[X_1] - p = p - p = 0$ unbiased

$\text{Var}(X_1) = p(1-p)$. When $p=0$ or 1 , this is good
worse when p is in between

$\text{Bias}(\bar{X}_n) = E[\bar{X}_n] - p = p - p = 0$ unbiased

$\text{Var}(\bar{X}_n) = \text{Var}(X_1)/n = \frac{p(1-p)}{n}$ good estimator
for n large, regardless of p .

Recall: if $E X_1 = \mu$, $\text{Var}(X_1) = \sigma^2$, then
 $E \bar{X}_n = \mu$, $\text{Var}(\bar{X}_n) = \sigma^2/n$



so $\text{Var}(\bar{X}_n) = \frac{1}{4n}$
in the worst case!

Def The mean-squared error (MSE) is

$$MSE(\hat{\theta}_n) = \mathbb{E}[(\hat{\theta}_n - \theta)^2]$$

Prop $MSE(\hat{\theta}_n) = \text{bias}(\hat{\theta}_n)^2 + \text{Var}(\hat{\theta}_n)$

Proof

$$\mathbb{E}[(\hat{\theta}_n - \theta)^2] = \mathbb{E}[\underbrace{(\hat{\theta}_n - \mathbb{E}\hat{\theta}_n)}_a + \underbrace{(\mathbb{E}\hat{\theta}_n - \theta)}_b]^2]$$

$$= \mathbb{E}[(\hat{\theta}_n - \mathbb{E}\hat{\theta}_n)^2 + 2(\hat{\theta}_n - \mathbb{E}\hat{\theta}_n)(\mathbb{E}\hat{\theta}_n - \theta) + (\mathbb{E}\hat{\theta}_n - \theta)^2]$$

$$= \mathbb{E}[(\hat{\theta}_n - \mathbb{E}\hat{\theta}_n)^2] \leftarrow \text{Var}(\hat{\theta}_n)$$

$$+ 2(\mathbb{E}\hat{\theta}_n - \theta) \underbrace{\mathbb{E}[\hat{\theta}_n - \mathbb{E}\hat{\theta}_n]}_{=0}$$

$$+ (\mathbb{E}\hat{\theta}_n - \theta)^2 \leftarrow \text{Bias}(\hat{\theta}_n)^2$$

$$= \text{Var}(\hat{\theta}_n) + \text{Bias}(\hat{\theta}_n)^2$$

Bias-variance tradeoff

hard to make both bias & variance small,
trade off between the two.

Def $\hat{\theta}_n$ is a consistent estimator if

$$MSE(\hat{\theta}_n) \rightarrow 0 \text{ as } n \rightarrow \infty$$

Back to example $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Ber}(p)$

$$\hat{p}_n = \frac{1}{2}, \quad \hat{p}_n = X_1, \quad \hat{p}_n = \bar{X}_n$$

$$MSE(\underline{\frac{1}{2}}) = \underline{(\frac{1}{2} - p)^2} + 0$$

NOT going to 0 as $n \rightarrow \infty$
so NOT consistent

$$MSE(X_1) = 0^2 + p(1-p)$$

NOT going to 0 as $n \rightarrow \infty$
so NOT consistent

$$MSE(\bar{X}_n) = 0^2 + \frac{p(1-p)}{n} \rightarrow 0 \text{ as } n \rightarrow \infty \checkmark$$

so it is consistent

